Problem 5.1, Stephens page 116

What's the difference between a component-based architecture and a service-oriented architecture?

In a component-based architecture, the system is built from loosely coupled components that run within the same executable program and communicate through direct method calls. On the other hand, a service-oriented architecture consists of independent services that run as separate programs and communicate over a network.

Problem 5.2, Stephens page 116

Suppose you're building a phone application that lets you play tic-tac-toe against a simple computer opponent. It will display high scores stored on the phone, not in an external database. Which architectures would be most appropriate and why?

Since no external database or network communication is required and the entire system runs on a single phone, a component-based architecture is most appropriate. All parts of the application such as UI, game logic, AI opponent, and high-score storage can run within the same executable and communicate through direct method calls.

Problem 5.4, Stephens page 116

Repeat question 3 [after thinking about it; it repeats question 2 for a chess game] assuming the chess program lets two users play against each other over an Internet connection.

Since two players are playing against each other over an Internet connection, a service-oriented architecture is more appropriate. The system would include a central server that runs as a separate service, and both client applications would communicate with it over the network. This allows the game state to be synchronized between the two players.

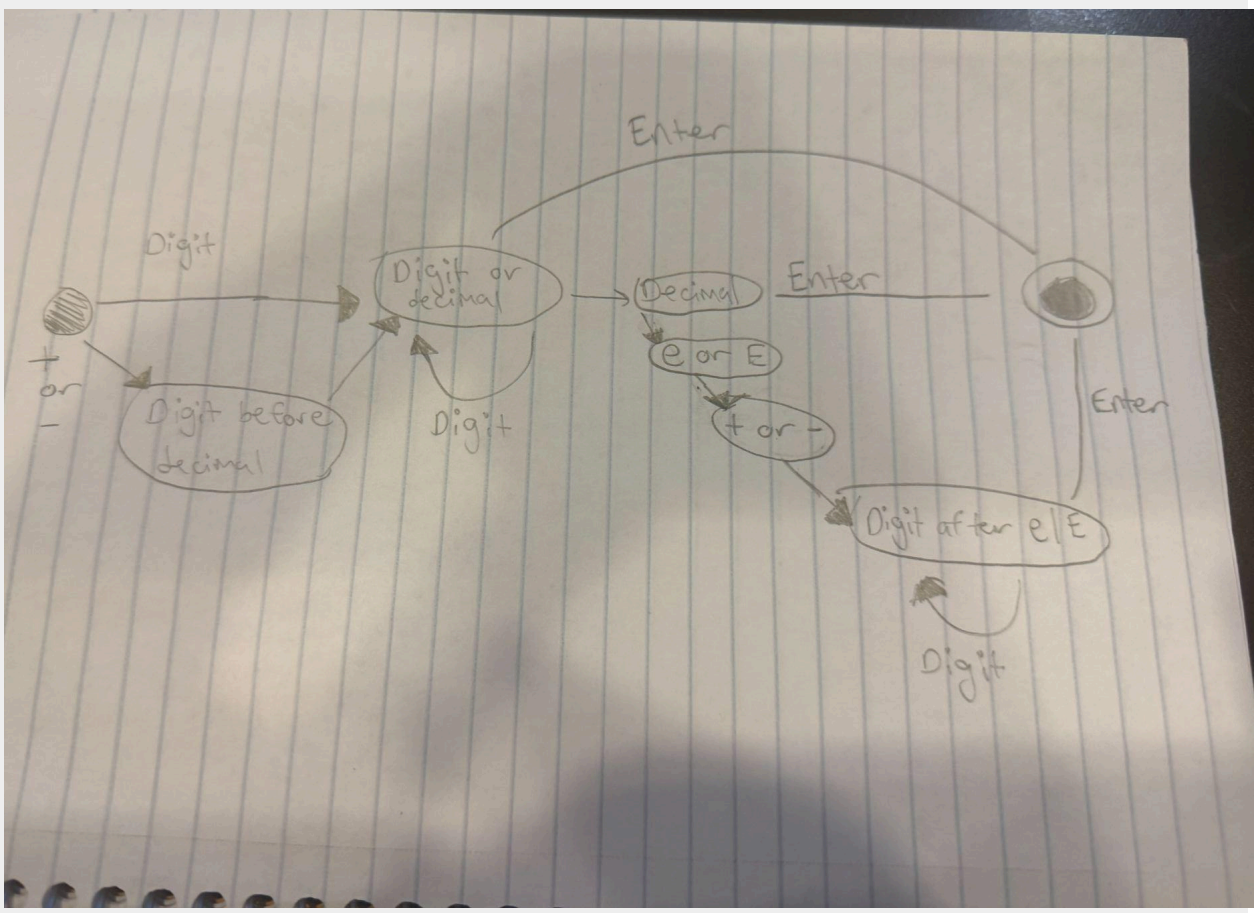
Problem 5.6, Stephens page 116

What kind of database structure and maintenance should the `ClassyDraw` application use?

The classy draw application should you the monolithic database structure

Problem 5.8, Stephens page 116

Draw a state machine diagram to let a program read floating point numbers in scientific notation as in +37 or -12.3e+17 (which means -12.3 x 10^{17}). Allow both E and e for the exponent symbol. [Jeez, is this like Dr. Dorin's DFAs, or *what*???]



Problem 6.1, Stephens page 138

Consider the `ClassyDraw` classes `Line`, `Rectangle`, `Ellipse`, `Star`, and `Text`.

1. What properties do these classes all share?

   All of the classes represent drawable objects, so they can be drawn, selected, moved, and reordered in the drawing.

2. What properties do they `NOT` share?

   Each class has different properties depending on its type, such as endpoints for a Line, dimensions for shapes, and text and font for Text.

3. Are there any properties shared by some classes and not others?

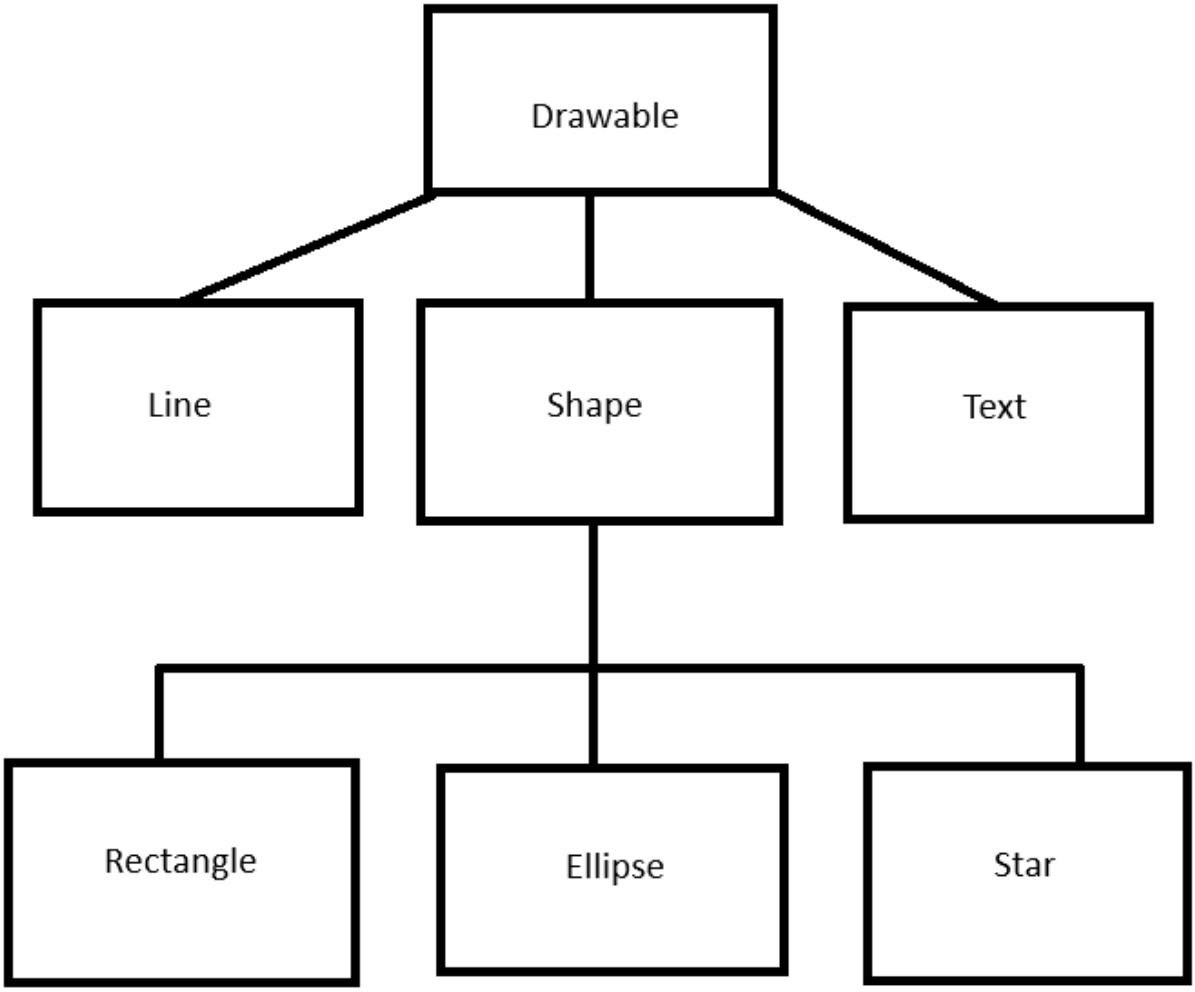
   Yes, some shapes share properties like width and height, while Text has font properties and Line has endpoints that the others do not share.

4. Where should the shared and nonshared properties be implemented?

   Properties shared by all classes should be in a parent class, properties shared by some classes should be in shared subclasses, and unique properties should be in each individual class.

Problem 6.2, Stephens page 138

Draw an inheritance diagram showing the properties you identified for Exercise 6.1. [Create parent classes as needed, and don't forget the `Drawable`class at the top.]


Drawable
Line
Shape
Text
Rectangle
Ellipse
Star